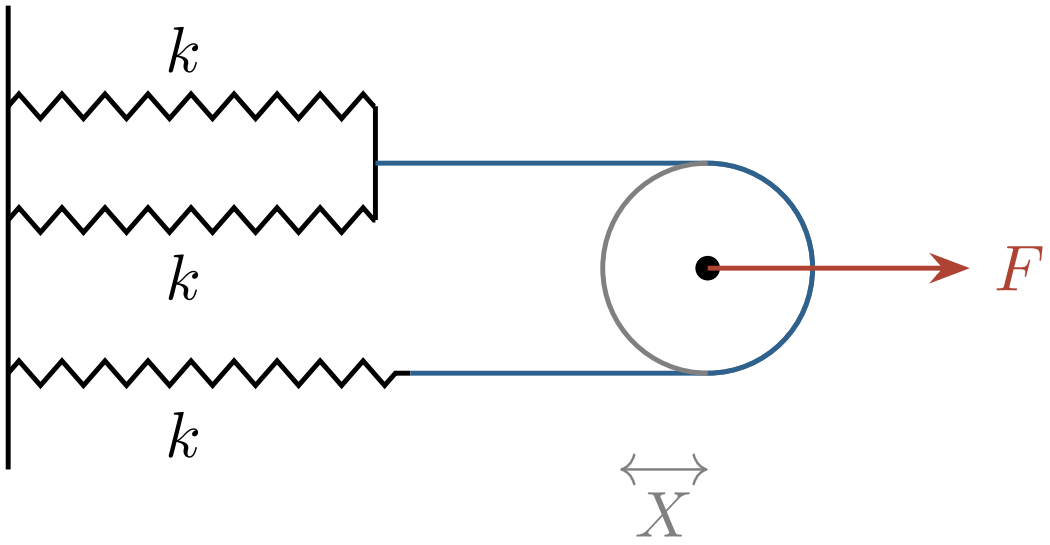
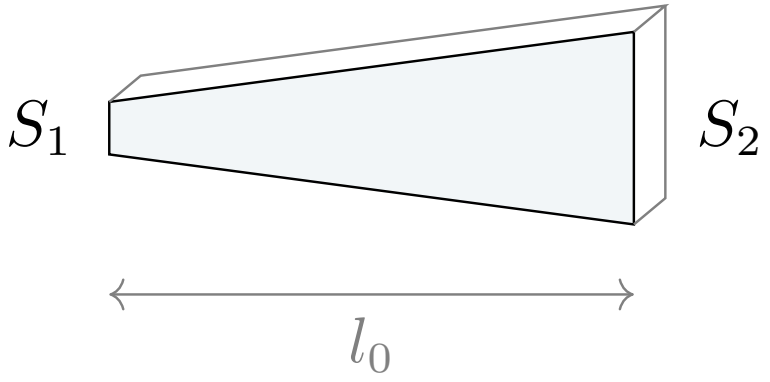
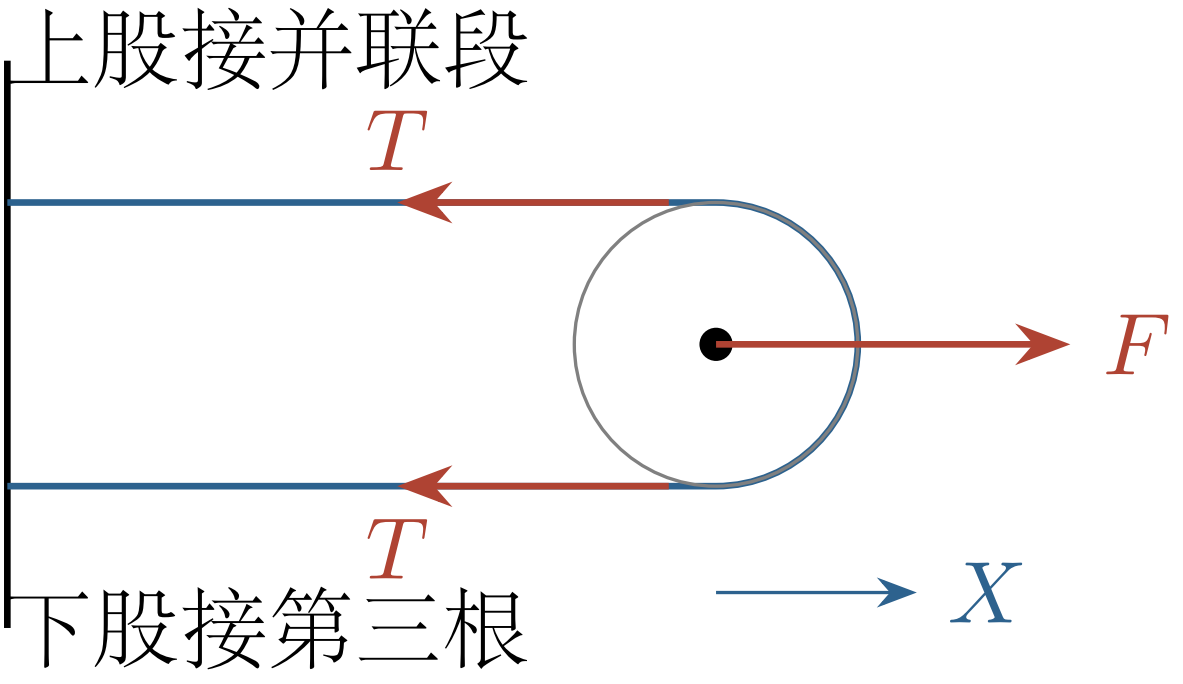


截面积沿长度线性变化

图示为恒厚度变宽条带

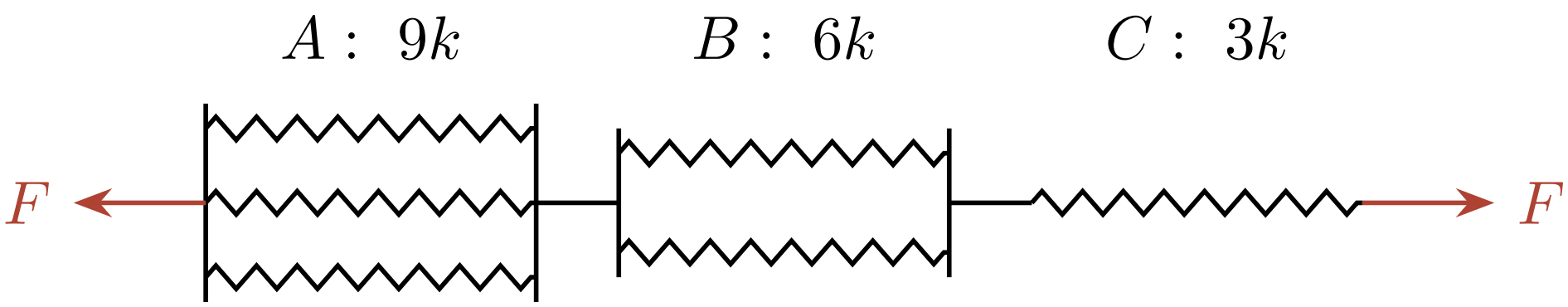


轮缘连接相切；并联组与第三根由同一轻绳相连



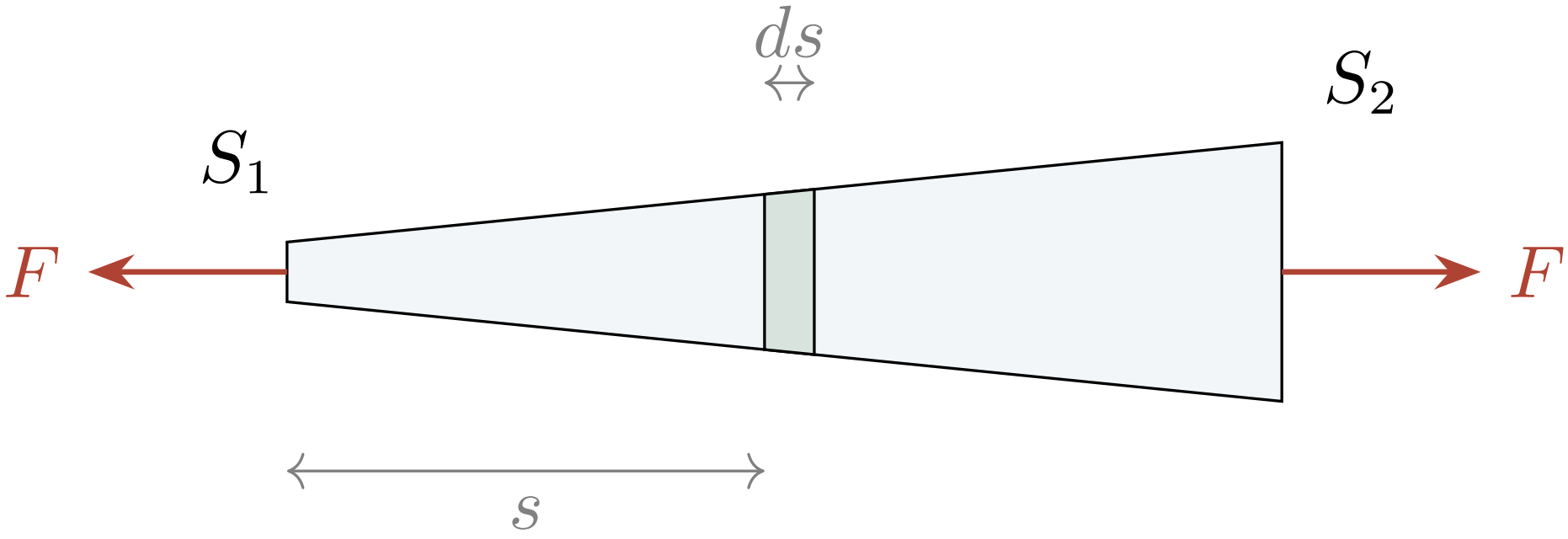
$$F = 2T$$

$$\delta_{\parallel} + \delta_3 = 2X$$



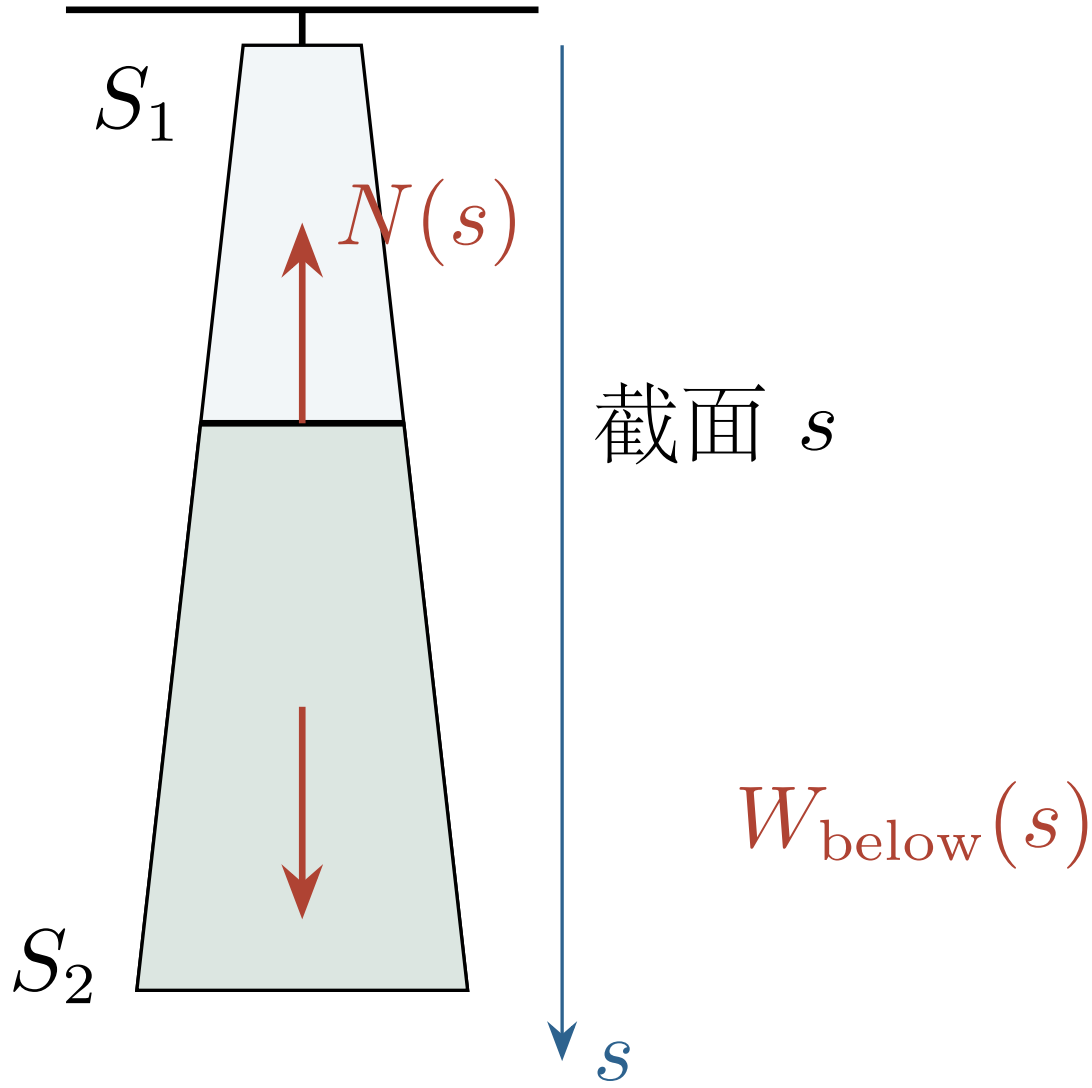
每根原长 $l_0/3$，单根刚度 $3k$

三段总拉力均为 F，总伸长逐段相加



$$S(s) = S_1 + (S_2 - S_1)s/l_0$$

$$d(\Delta l) = F\,ds/[ES(s)]$$



$$W_{\text{below}}(s) = \rho g \int_s^{l_0} S(\xi) d\xi$$

自由端 $N(l_0) = 0$；截面拉力等于阴影段重量